

① Apriori Algorithm - online shopping Transactions.

pseudocode

1. Find support count of all 1-itemsets.
2. Remove itemsets whose support $<$ min-support
3. let $L_1 =$ Frequent 1-itemsets.
4. let $L_1 =$ frequent 1-itemsets.

$C_k =$ Generate candidates (L_{k-1})

for each transaction T in D :

count support of candidates in C_k that are
in T .

$L_k =$ candidates in C_k having support $\geq$ min-
support.

if L_k is empty:

stop.

so return all frequent itemsets.

② FP - Growth - library Book Borrowing Records.

Pseudocode :-

1. Scan D and calculate support of each item.
2. Remove items below minimum support.
3. Sort remaining items in descending support order.
4. Create an empty FP-Tree.
5. For each transaction T:
 - Sort items according to global frequency.
6. Create a header table linking identical items.
7. For each item i:
 - Find its conditional FP-Tree.
8. Construct its conditional FP-Tree.
- Generate frequent patterns recursively.
9. Return all frequent itemsets.

③ Association Rule Generation - frequently purchased services.

pseudocode :-

1. for each frequent itemset L in F
2. Generate all non-empty proper subsets of L .
3. for each subset A :
4. $B = L - A$.
5. confidence = support(L) / support(A).
6. if confidence $\geq$ min-confidence.
7. output rule $A \rightarrow B$.

④ constraint-based frequent itemset mining student courses.

pseudocode :-

1. select only item sets containing "python"
2. Generate candidate 1-itemset :
3. check its support.

4. if $\text{support} \geq \text{min-support}$:

keep {python}.

5. generate larger candidates by adding other courses:

6. count their support:

7. prune candidates that:

a) do not contain python

8. generate further candidates only from remaining.

9. return frequent itemsets containing python.

⑤ multilevel Association Rule Mining - Hospital Treatment Hierarchy.

1. start from the highest hierarchy level.

2. find frequent treatment itemsets.

3. generate association rules at this level.

4. move to the next lower hierarchy level.

5. replace general treatments with their specific.

6. count supports and prune infrequent itemsets

7. repeat until the lowest hierarchy level is reached.

8. output rules from all levels.